$$\sigma_Q(a \mid s) = \frac{e^{Q(s,a)}}{\sum_b e^{Q(s,b)}}$$

$$c_H(s) = \log \sum_b e^{Q(s,b)} - \sum_b \widehat{\pi}_\beta(b \mid s) Q(s,b)$$

$$\frac{\partial c_H(s)}{\partial Q(s,a)} = \sigma_Q(a \mid s) - \widehat{\pi}_\beta(a \mid s)$$